

Prediction of Human Mobility: An Approach to Predicting Next Location Patterns from Past Movement History

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Abstract

This report addresses the challenge of human mobility prediction using large scale movement data from mobile devices. The primary goal is to accurately predict the next location of an individual based on their historical movement patterns. This task is inherently difficult due to several factors: significant variations in travel patterns across different cities, the unique routines and behaviors of each user (user diversity), and the prevalence of incomplete data, which includes gaps, missing observations, and irregular sampling intervals. We explore a range of models, from statistical baselines to more complex pattern matching and nearest neighbor approaches, to determine the most effective strategies for this realistic and challenging dataset.

1 Introduction: Motivating Use Cases

The ability to accurately forecast human movement patterns from large scale data is no longer just an academic challenge; it has profound practical applications across critical sectors of society. Predicting human mobility has several critical applications in various domains.

- **Urban Planning and Transportation:** By forecasting how and where large groups of people will move, city planners can optimize bus/train routes and allocate transit resources more efficiently.
- **Traffic Management and Congestion Control:** Authorities can reduce congestion by anticipating peak hour surges, which enables the implementation of smarter traffic signals and dynamic routing for drivers.
- **Public Health and Epidemic Modeling:** During a public health crisis, mobility models can estimate large scale contact rates, helping to plan more effective responses and targeted lockdowns to model disease spread.

This report details a systematic exploration of various models to tackle this prediction task, navigating the common challenges of real world mobility data.

2 Literature Review

Recent advancements in the Human Mobility Prediction Challenge (HuMob) have demonstrated a divergence in successful methodologies, ranging from sophisticated deep learning architectures to refined classical machine learning and probabilistic approaches. This section reviews five key contributions that address the challenges of data scarcity, cross city generalization, and irregular temporal patterns.

Addressing the challenge of predicting mobility in data scarce cities, Terashima et al. [3] proposed a multi city transfer learning framework using multiple BERT models. Their approach utilizes Time-series Stay Frequency (SF-BERT) to capture abstract, city independent stay tendencies and City Movement (CM-BERT) to learn city specific spatial distributions. By combining these encodings into a Location Prediction BERT (LP-BERT), they successfully transferred knowledge from data rich cities to those with fewer users. Similarly tackling the complexity of long term cross city prediction, He et al. [4] introduced ST-MoE-BERT, which integrates a Mixture-of-Experts (MoE) architecture with a BERT backbone. Their MoE layer employs multiple specialized expert networks to model distinct spatio-temporal dynamics, while a transfer learning strategy with differential learning rates allows the model to adapt quickly to new cities.

In contrast to deep learning approaches, Imai et al. [5] demonstrated that classical machine learning can remain competitive through rigorous feature engineering. By employing Support Vector Regression (SVR) with a non-linear RBF kernel and introducing novel features such as movement counts and average travel angles alongside standard temporal variables, they achieved a significant improvement in Dynamic Time Warping (DTW) scores over rule based baselines.

Focusing on the cyclical nature of human behavior, Ishikawa [2] proposed a probabilistic improvement to the simple weekly rhythm model. Instead of assuming a deterministic 7-day cycle, this method calculates the probability $P(dow|d)$ that a specific date d behaves like a certain day of the week dow . This allows the model to flexibly handle irregularities, such as a Monday holiday exhibiting behavioral patterns more similar to a Sunday.

Finally, Li et al. [1] introduced a temporal-trajectory-based K-nearest neighbor (TT-KNN) algorithm that is specifically fine tuned based on three fundamental characteristics of human mobility: time fixedness, contingency, and uniqueness. Their approach involves an offline process of cleaning incidental locations and segmenting trajectories by time, followed by an online prediction phase using a hashmap based retrieval system. This method effectively addresses issues of data imbalance and noise, demonstrating that models grounded in mobility theory can outperform generic deep learning baselines like GPT in terms of spatial accuracy.

3 Problem Statement

The core objective of this project is to predict the next location of each person based on their past movement history, using large scale movement data from mobile devices.

This task is particularly challenging due to the nature of the data and human behavior:

- **Different Cities:** Travel patterns, routines, and popular locations vary widely across different metropolitan areas. A model trained on one city may not generalize well to another.
- **User diversity:** Each person has unique routines (e.g., a fixed work commute) and mobility behaviors (e.g., spontaneous travel).

- **Incomplete data:** Real world data is imperfect. The dataset contains gaps, missing observations, and irregular sampling, which makes it difficult to build a complete picture of a user’s trajectory. This is a realistic constraint, and the goal is to build a model robust enough to handle it.

4 Data and Methodology

4.1 Data Sources

The data used for this originates from the **ACM SIGSPATIAL GIS Cup 2025 challenge**. It provides human mobility patterns for four distinct metropolitan areas, anonymized as Cities A, B, C, and D. These trajectories were generated from real world smartphone application location data provided by LY Corporation.

4.2 Dataset Layout

The dataset is structured as a time series of location visits, with each record containing (uid, date, time, x, y), as shown in Figure 1.

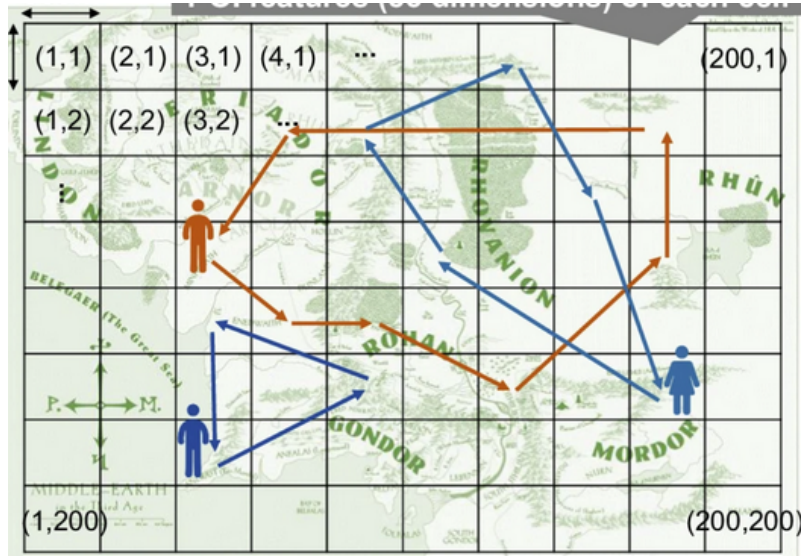


Figure 1: Data structure [8]

- **Training Window:** Movement data is provided for days 1-60 for all individuals in each city.
- **Prediction Target:** The goal is to predict the movements of 3000 specific target users per city during a future time window.
- **Masked Data:** For these target users, all movements from days 61-75 are masked (with the location encoded as “999”).
- **Temporal Gaps:** A key challenge is introduced in Cities C & D, where the prediction window (days 61-75) occurs several months after the training window, unlike in Cities A & B where it immediately follows.

4.3 Dataset Statistics

As shown in Table 1, the dataset is large scale, with City A alone comprising 150,000 users and 87 million trajectory records. This scale presents significant computational challenges.

Table 1: Dataset Statistics

City	Total Users	Rows	Target Users	Avg Trajectory Length	Masked Rows
A	150000	87 M	3000	427	320 K
B	30000	18 M	3000	505	370 K
C	25000	14 M	3000	502	290 K
D	20000	12 M	3000	522	310 K

5 Algorithm and Internal Working: Approaches Tried

We implemented and evaluated a series of models, progressively increasing in complexity from simple statistical baselines to advanced pattern matching techniques.

5.1 Statistical Baselines

A set of statistical baselines were established to measure the baseline performance and provide a point of comparison.

5.1.1 Global Aggregates

These models ignore individual user behavior and make a single prediction for all users.

- **Global Mean:** Predict all user’s future staypoints as the average location (centroid) of all data from days 1-60.
- **Global Mode:** Predict all user’s future staypoints as the single most frequent location visited by all users in days 1-60.

5.1.2 User-Specific Aggregates

These models create a unique prediction for each user but do not capture sequential or temporal patterns.

- **Per-User Mean:** For each user, predict their future staypoints as their personal average location from days 1-60.
- **Per-User Mode:** For each user, predict their future staypoints as their personal most frequent location from days 1-60.

5.1.3 Basic Language Model (N-Gram)

To capture simple sequential patterns, we adapted n-gram language models to location data.

- **Unigram Model:** Predicts a user’s next location by sampling from their personal distribution of location frequencies (e.g., 70% home, 30% office $\rightarrow$ prediction follows these ratios).
- **Bigram Model:** Models the probability of transitions (e.g., $P(loc_B|loc_A)$). It uses the frequency of pairs of consecutive locations to capture routines like home $\rightarrow$ office.
- **Bigram + Top-p (0.7):** A variation of the Bigram model that samples only from the most likely 70% of transitions, adding variation and avoiding overly deterministic predictions.

5.2 Topological Temporal k-Nearest Neighbors (TT-KNN)

Overview Based on the approach by Li et al [1], this model predicts the next location of a user by finding similar past transitions in their own trajectory.

Key steps The model’s logic is built on pre processing the trajectories into a lookup index:

- **Filtering:** Ignore rare locations (visited $< \tau$ times, default $\tau = 5$) to reduce noise.
- **Segments:** Split time into fixed intervals ($\Delta = 30$ mins).
- **Transitions:** Record moves between locations that occur in consecutive time segments.
- **Indexing:** Build a hashmap (TT-index) that maps (time segment, current location) $\rightarrow$ {list of possible next locations}.

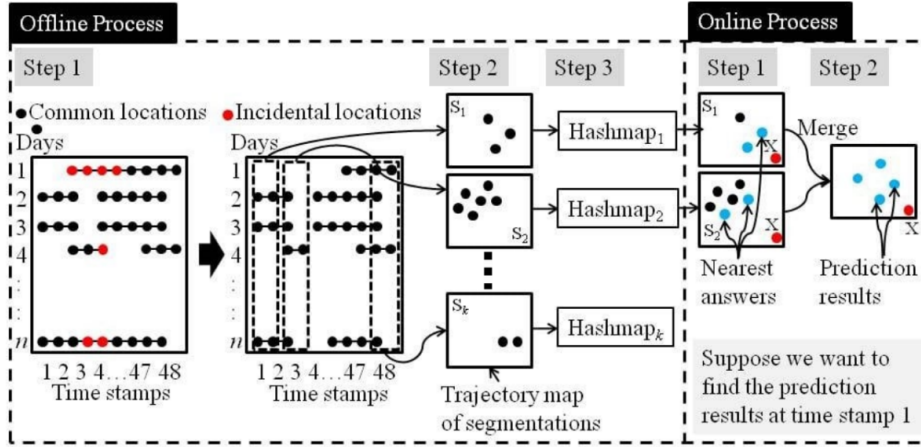


Figure 2: Overview of the TTKNN model architecture. [1]

Training and Inference

- **Training:** Parse each user’s trajectory (days 1-60) and build their personal TT-index.
- **Inference:** For each masked test point, find the nearest (in time) transitions in the TT-index and predict the next location using k-nearest neighbors logic. If no match is found, it falls back to predicting the user’s current location.

5.3 Frequency TT-KNN

Problem in TT-KNN: The standard TT-KNN model treats all candidate next locations equally if they appear in the TT-index, ignoring how often a particular transition actually happens.

Key Improvement: We introduced a frequency weight to the scoring logic:

- Track the count for each transition ($loc_1 \rightarrow loc_2$).
- Combine distance and frequency into a single weighted score: $\text{score} = \text{distance} / (\text{frequency weight})$.
- This prioritizes transitions that are both temporally close and historically frequent.

5.4 Clustered TT-KNN

Why clustering? Users exhibit very different movement patterns (e.g., sparse vs. dense trajectories). For users with sparse data, their individual TT-KNN index is weak. Grouping similar users allows them to share trajectory knowledge.

Key improvements to TT-KNN:

- Represent each user as a vector based on their visited cells.
- Apply clustering methods (K-Means, BIRCH, HDBSCAN) to group similar users.
- Build cluster level transition tables and hourly profiles.
- Use a hybrid prediction: first, try the user’s personal TT-KNN. If data is weak (few transitions), fall back to the patterns of their assigned cluster.

5.5 Predictive Pattern Matching (PPM)

As an alternative to transition-based models, we explored PPM, which attempts to match the entire day’s partially known trajectory to historical day patterns (see Figure 3).

5.5.1 Strategy 1: Cosine Similarity

How it works: This strategy scores the similarity between the current day (with masked entries) and a historical day.

- Compare the current day’s revealed trajectory with each historical day.
- Count how many timestamps have exact location matches.
- $\text{Score} = (\text{matching locations}) / (\text{total common timestamps})$.

Prediction Process: Select the top 10 most similar historical days, and for each masked timestamp, predict the location by a majority vote across those 10 days.

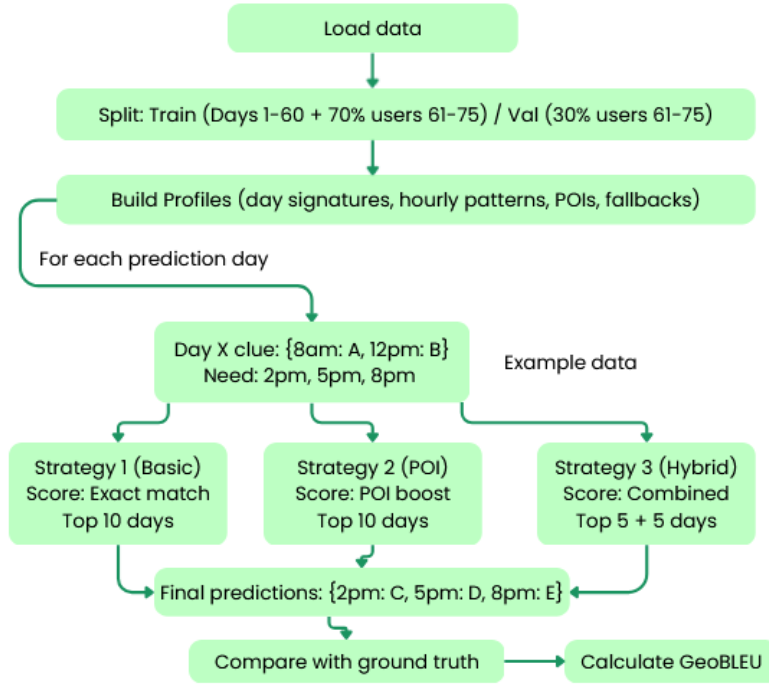


Figure 3: Overview of the PPM model pipeline.

5.5.2 Strategy 2: POI-Aware

Key Difference: This strategy posits that matches at popular locations (Points of Interest, or POIs) are more significant than matches at obscure locations.

Enhanced Scoring:

- Exact POI match: +2.0 points (double weight).
- Regular match: +1.0 point (standard).
- Both are POIs but no match: +0.5 points (partial credit for similar activity type).

POI-Based Fallback: If no similar days are found, the model falls back to predicting the most popular global POI for that specific timestamp.

5.5.3 Strategy 3: Hybrid Approach

Approach: This combines the POI focused (high precision) and regular (broad coverage) strategies.

Two Phase Candidate Selection:

- Step 1: Run POI-aware matching → Get 5 best days.
- Step 2: Run basic matching → Get 5 best different days.
- Step 3: Merge into a single pool of 10 candidates.

- Step 4: Perform the majority vote across all 10 candidates to make the final prediction.

6 Metric: GeoBLEU

6.1 What is GeoBLEU?

The primary evaluation metric for this challenge is GeoBLEU. It is a geospatial adaptation of the BLEU (Bilingual Evaluation Understudy) score, which is traditionally used for machine translation. It is designed to compare a predicted trajectory (candidate) against an actual trajectory (reference) as sequences of locations.

Instead of requiring exact n-gram (sequence of locations) matches, GeoBLEU allows for spatial proximity. Nearby grid cells can partially match, with the score decaying exponentially as the distance between the candidate and reference location increases.

6.2 How it works (high level):

- Break each trajectory into overlapping geospatial n-grams (windows of successive locations).
- For each candidate n-gram, match it to the closest unused reference n-gram.
- The score for a match is calculated as $\exp(-\text{distance})$.
- A modified precision score is computed over all matched n-grams, which forms the final GeoBLEU score.

6.3 Interpretation:

- A score of 1.0 indicates a perfect match in both sequence and spatial location.
- Lower scores indicate divergence in either the order of visits, the specific locations, or both.
- This metric is effective for evaluating how well a model captures the local route structure, not just the final destination.

7 Results, Observations, Inferences

All models were evaluated using the GeoBLEU metric (detailed in Section 6).

7.1 Statistical Baseline Results

Table 2 presents the performance of the statistical baselines. As expected, global models perform very poorly. The user-specific models show a significant improvement, with Per-User Mode and Bigram Model (top_p=0.7) establishing a non trivial baseline score of ~ 0.084 .

Table 2: Statistical Baseline Results (GeoBLEU Score)

Model	City A	City B	City C	City D	Average
Global Mean	0.00029	0.00025	0.00107	0.00001	0.00040
Global Mode	0.00126	0.00419	0.00493	0.00191	0.00308
Per-User Mean	0.01651	0.01778	0.01588	0.02026	0.01761
Per-User Mode	0.07918	0.08420	0.08338	0.08866	0.08385
Unigram Model	0.03585	0.04105	0.03726	0.04309	0.03932
Bigram Model	0.05243	0.05374	0.05380	0.06066	0.05666
Bigram Model (top-p=0.7)	0.07819	0.08984	0.08002	0.09044	0.08463

7.2 TT-KNN Model Results

The TT-KNN models, detailed in Table 3, demonstrated steady improvement. The basic TT-KNN (0.09183) already outperforms the best baseline. Adding frequency weighting (0.09768) provides a consistent boost. Finally, clustering users before applying TT-KNN yields the best results in this category, with HDBSCAN achieving the highest average score (0.10438), confirming the hypothesis that sharing knowledge among similar users is beneficial.

Table 3: TT-KNN Model Results (GeoBLEU Score)

Model	City A	City B	City C	City D	Average
TT-KNN	0.08741	0.09649	0.08558	0.09782	0.09183
Frequency TT-KNN	0.09603	0.10185	0.09032	0.10250	0.09768
Cluster TT-KNN (K-Means)	0.10480	0.10626	0.09532	0.10638	0.10319
Cluster TT-KNN (BIRCH)	0.10532	0.10679	0.09621	0.10660	0.10373
Cluster TT-KNN (HDBSCAN)	0.10572	0.10724	0.09708	0.10748	0.10438

7.3 PPM Model Results

Table 4 shows the results for the PPM strategies. Surprisingly, the simplest Cosine Matching (No POI) strategy achieved the highest average score of all models tested (0.105180). The more complex POI-Aware and Hybrid models did not improve upon this, suggesting that simple trajectory matching was the most robust strategy.

Table 4: PPM Model Results (GeoBLEU Score)

Model	City A	City B	City C	City D	Average
Cosine Matching (No POI)	0.102656	0.110432	0.101560	0.106077	0.105180
Cosine Matching (POI-Aware)	0.102500	0.110233	0.101368	0.105866	0.104992
Hybrid Model	0.098718	0.106067	0.097953	0.102318	0.101264

7.4 Inferences (Key Takeaways)

- Statistical models like Top-p bigram models showed good performance for their simplicity.

- TTKNN models demonstrated steady improvement over different versions, with clustering providing a significant boost.
- PPM models outperformed the TTKNN family, despite being conceptually simpler.
- The basic PPM (Cosine Matching) version surprisingly achieved the highest overall score among all tested models, indicating that direct daily pattern matching was a very effective strategy.

8 Challenges and Future Improvements

8.1 Failed Attempts

During our project, several advanced models were explored but proved unsuitable given their constraints.

- **Deep learning models:** We attempted to use a basic LSTM network. However, it was computationally prohibitive, taking over 10 hours for just 5 epochs and yielding a very low score. The large dataset and limited computational resources were significant bottlenecks.
- **Time series models:** We also attempted a zero shot approach with TimesFM [6]. The score was almost zero, as the model does not have inherent spatial trajectory awareness and is not designed to handle categorical location data.

8.2 Challenges Encountered

- **User diversity:** Routines vary widely, from rigid daily commutes to highly random patterns. Modeling this diversity in a single framework is difficult.
- **Large scale data:** The sheer volume of data (87M rows for City A) led to high training times and frequent kernel crashes.
- **Incomplete data:** The data gaps, especially the multi month gap in Cities C and D, make it difficult for models to learn stable patterns.
- **Metric limits:** The GeoBLEU metric, being implemented in NumPy, was very slow to compute for large cities like City A.

8.3 Further Improvements

Based on our findings, two primary future work are proposed:

- **Deep Learning Models:** Develop a more sophisticated training strategy for deep learning models, such as using sub sampling or more efficient data loaders, to reduce training time and improve scores.
- **Transfer Learning:** Explore strategies to reuse models trained on data rich cities (like City A) to improve performance on other cities with minimal retraining.

9 Conclusion

This project conducted a comprehensive evaluation of multiple models for human mobility prediction on a large, realistic, and challenging dataset. The results indicate that while advanced, clustered TT-KNN models show consistent improvement over baselines, a simpler **Predictive Pattern Matching (PPM)** model based on cosine similarity achieved the highest overall score. This surprisingly strong performance suggests that for this dataset, matching daily patterns as a whole was a more effective strategy than modeling discrete, step by step transitions.

10 References

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